

SIT720 Machine Learning Mini research - HD Task

Pranjal Gupta S226230386

September 2026

GitHub repository (code, notebooks, data, README): <https://github.com/gpariee/HD-Machine-Learning-Mini-Project>

Video presentation link: <https://drive.google.com/file/d/1HLwDioGNV-u2mArWC496cAxhsx5nvSmZ/view?usp=drivesdk>

Report topic: Reproduction and Extension of a Stacking-Based Ensemble for Early Heart Attack Prediction (Technical Research Report – Parts 1 & 2)

Executive Summary

This report reproduces and critically extends **Bhagat, Sharma & Agarwal (2025)**, “*An efficient stacking-based ensemble technique for early heart attack prediction*,” published in *Multimedia Tools and Applications* [1]. The paper reports a six-classifier stacking ensemble (Logistic Regression, Decision Tree, Random Forest, XGBoost, Naive Bayes, K-Nearest Neighbors) that achieves 98.53% accuracy on a 1025-patient heart disease dataset.

Part 1 faithfully reproduces this pipeline. In doing so we identify that the dataset file used by the paper contains **723 of 1025 rows (70.5%) as exact duplicates**, and that de-duplication leaves only 302 unique rows – essentially the original, single-source UCI Cleveland dataset (n=302/303), not the four-hospital merge the paper describes. We show, by holding the modelling pipeline fixed and varying only whether duplicates are removed before splitting, that this single change explains the overwhelming majority of the gap between our reproduction and the paper’s reported near-perfect performance: our stacking ensemble scores **99.40% +/- 0.74% accuracy** on the leaky, as-published data (closely matching the paper’s 98.53%) but only **83.11% +/- 5.29%** accuracy once duplicates are removed.

Part 2 proposes and evaluates two complementary, substantially different methodological contributions, both addressing limitations exposed in Part 1:

1. A **leakage-safe, nested-cross-validated, feature-engineered and hyperparameter-optimised pipeline**, evaluated with a paired Wilcoxon signed-rank test against an untuned baseline on identical folds. We report, honestly, that on this small (n=302) de-duplicated dataset the additional engineering does **not** yield a statistically significant improvement (all $p > 0.05$, though precision comes close at $p=0.055$) – itself an important, evidence-based finding about the limits of what further modelling sophistication can achieve once the leakage bias is removed.
2. A **genuine four-hospital merge and cross-hospital (leave-one-site- out) generalisation study**, built from the four original raw UCI site files. This reveals that two of the paper’s own top-ranked predictive features (**ca**, **thal**) are missing at 42-99% of patients outside Cleveland – **ca** alone is 96-99% missing at every non-Cleveland site – and that cross-hospital AUC falls to 0.72-0.76 for Switzerland/VA versus 0.89 for a pooled random split – a real, clinically relevant generalisation gap the original paper’s protocol cannot detect.

All code, notebooks, data and results referenced in this report are included in the accompanying archive (see Appendix C for the README and reproduction instructions).

1 Introduction

1.1 Selection of the paper for reproduction

Three candidate papers were considered for this assignment:

1. Bhagat et al. – a DASMcC ECG-signal-feature multi-class CVD classifier on the public PhysioNet PTB-XL dataset [related IEEE Access paper, Option 1].
2. A cascading ensemble for annotating open IoT sensor metadata (Option 2, IEEE Internet of Things Journal).
3. **Bhagat, Sharma & Agarwal (2025)**, a stacking ensemble for heart attack prediction on a public tabular UCI-derived dataset (Option 3, *Multimedia Tools and Applications*) [1].

Option 2 was ruled out first: it relies on bespoke, non-public scraped IoT metadata with no clearly specified acquisition or preprocessing protocol, making faithful reproduction infeasible within the scope of this assignment. Option 1 uses a genuinely public dataset (PTB-XL) but requires a heavier, more failure-prone signal-processing pipeline (peak-detection feature extraction across ~21,800 raw 12-lead ECG records). Option 3 was selected because (a) its dataset is small, fully public, and – as this report demonstrates – independently verifiable against the paper’s own reported feature ranges; (b) its methodology (six classical classifiers plus a stacking ensemble) is standard and fully within reach of a rigorous, faithful reproduction; and (c), as detailed below, it exhibits a data-quality issue rich enough to support a substantive, evidence-based critical analysis and a genuinely novel Part-2 contribution.

1.2 Report structure

Section 2 covers Part 1: the research problem, dataset, methodology, results and critical analysis of the reproduction. Section 3 covers Part 2: the proposed solution(s), their motivation, methodology, results and comparative analysis. Section 4 concludes. References follow in IEEE numbered style. Appendices contain supplementary tables, the assumption log, and reproducibility instructions.

2 Part 1 – Reproduction

2.1 Research problem, motivation, state of the art, and research gap

Research problem. Cardiovascular disease is a leading global cause of death, and early risk prediction from routinely collected clinical attributes (age, blood pressure, cholesterol, ECG findings, etc.) can support earlier intervention. The paper frames this as a binary classification problem: given 13 clinical features, predict presence (1) or absence (0) of heart disease.

Motivation. The authors motivate their approach by observing that individual classifiers plateau in the low-to-mid 90s (percent accuracy) on this class of dataset, and propose that a **stacking ensemble** – training a meta-classifier on the out-of-fold predictions of several heterogeneous base learners – can push performance further by combining each base learner’s complementary strengths.

State of the art referenced by the paper. The paper situates itself against a body of prior work on UCI-derived heart-disease datasets, citing accuracies ranging from roughly 85% to 99% across different combinations of feature selection (Lasso, chi-square/PCA, Relief, fast conditional mutual information) and classifiers (SVM, hybrid random-forest-linear-model ensembles such as Mohan et al.’s HRFLM [2], majority-voting ensembles, and others). The claimed contribution is that a five-fold stacking ensemble of six classical learners exceeds all of these at 98.53% accuracy.

Results reported in the paper (Table 11, reproduced in Section 2.4 below). Individual classifiers range from 84.4% (LR, NB) to 92.7% (DT, RF) accuracy; the stacking ensemble reports 98.53% accuracy, 100% precision, 97.27% recall, 98.61% F1 and 98.8% AUC.

Research gap addressed by the paper (as stated by the authors). The authors position their gap as: prior individual-classifier studies on this dataset plateau below 93% accuracy, and no prior work (in their citation list) had applied a 5-fold stacking ensemble of exactly this six-learner combination to this dataset.

Research gap this report addresses (identified during reproduction). As detailed in Section 2.5, our reproduction reveals a different, more fundamental gap: the paper does not verify that its train/test split is free of duplicate-patient leakage, and – independently – never tests whether its “four-database merge” claim is consistent with the very high missingness that a genuine merge of the four named UCI hospital sites would necessarily contain. Both gaps are addressed in Part 2.

2.2 Dataset and feature set

Dataset used by the paper. The paper states (Sec. 4) that the dataset “is made up of four databases: Cleveland, Hungary, Switzerland, and Long Beach V,” comprising 1025 patient records and 14 columns (13 features + binary target), reproduced verbatim in the paper’s Table 9.

Dataset used for our reproduction. The paper does not give a direct download link or DOI for its exact file. We sourced the file from a public GitHub mirror of the well-known Kaggle “Heart Disease Dataset” (uploader: johnsmith88) and verified it is the correct file by checking every numeric feature range the paper reports in its own Table 9. Source data for the comparison below: `results/part1/eda_summary.json`.

Dataset verification – our file vs. the paper’s Table 9 (all five reported ranges match exactly):

Feature	Paper’s reported range (Table 9)	Range in our file	Match
age	29-77	29.0-77.0	Yes
trestbps	94-200	94.0-200.0	Yes
chol	126-564	126.0-564.0	Yes
thalach	71-202	71.0-202.0	Yes
oldpeak	0-6.2	0.0-6.2	Yes

The file has 1025 rows, 14 columns, no missing values, and a near-balanced target (526 positive, 499 negative), consistent with the paper’s description.

Feature set (13 predictors + 1 target), as specified by the paper’s Table 9 and used identically here: `age`, `sex`, `cp` (chest pain type, 0-3), `trestbps` (resting blood pressure), `chol` (serum cholesterol), `fb`s (fasting blood sugar > 120mg/dl), `restecg` (resting ECG results, 0-2), `thalach` (max. heart rate achieved), `exang` (exercise-induced angina), `oldpeak` (ST depression), `slope` (slope of peak exercise ST segment, 0-2), `ca` (number of major vessels, 0-3), `thal` (1=normal, 2=fixed defect, 3=reversible defect); target: 1 = disease, 0 = no disease.

No feature engineering, feature selection, or dimensionality reduction is described in the paper beyond “preprocessing” (missing-value handling, scaling, categorical encoding) – all thirteen raw features are used directly, which we replicate in Part 1.

2.3 Machine learning methods and experimental protocol

Methods (Table 8 / Sec. 3.3 of the paper). Six base classifiers – Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), XGBoost (XGB), Gaussian Naive Bayes (NB), and K-Nearest Neighbors (KNN) – are each trained independently, then combined via a **5-fold stacking ensemble**: a meta-classifier is trained on the out-of-fold predictions of the six base learners, following the standard stacking recipe (the paper’s Table 8 algorithm box gives this five-step procedure but no further detail).

Experimental protocol described by the paper. “For all input dataset, split into training and testing dataset” -> “Preprocess both training and testing data” -> “Apply all six classifiers ... to make initial prediction over training dataset” -> “Stack all these classifiers to build meta classifier” -> “Evaluate model performance.” This is the entirety of the protocol description; the paper gives **no train/test split ratio, no random seed, no base-learner hyperparameters, and no meta-classifier identity**.

Assumptions made to enable reproduction (justified below; also logged in Appendix A of this report and in the docstring of `src/part1_reproduction.py`):

- **A1 (split ratio):** an 80/20 stratified train/test split is assumed, the field-standard default and the same ratio explicitly reported for the companion Option-1 ECG paper considered for this assignment.
- **A2 (random seed / stochastic variation):** `random_state=42` is fixed throughout; in addition, because the paper reports only a single run, we additionally report results averaged over 30 repeated random stratified splits, to characterise how much of the paper’s reported number could plausibly be sampling luck versus a stable estimate.
- **A3 (base-learner hyperparameters):** scikit-learn / XGBoost defaults are used for all six base learners, since the paper specifies none.

- **A4 (meta-classifier identity):** Logistic Regression is used as the stacking meta-classifier – the conventional default choice for a classification stack and consistent with the paper’s own textual description of “a trained meta-classifier.”
- **A5 (scaling):** `StandardScaler`, fit on the training fold only, is applied to all thirteen features before every base learner, consistent with the paper’s statement that data is “scale[d] ... correctly” but otherwise unspecified.

Two experimental protocols are run in parallel (not both faithful reproductions of the paper – Protocol A is; Protocol B is a diagnostic control described further in Section 2.5):

- **Protocol A (“as-published”):** the full 1025-row file, ordinary random stratified 80/20 split – our best-effort faithful reproduction of the paper’s stated procedure.
- **Protocol B (“de-duplicated”):** identical pipeline and code, but exact duplicate rows are removed before splitting (leaving 302 unique rows). This isolates the causal effect of duplicate-row leakage, holding every other modelling choice fixed.

2.4 Evaluation metrics and experimental results

Metrics (as specified by the assignment brief and matching the paper’s Table 10): Accuracy, Precision, Recall, F1-score, and AUC. We additionally record Specificity and the Matthews Correlation Coefficient (MCC) – both reported by the paper, and MCC in particular recommended in the literature as a more informative single summary statistic than accuracy or F1 for binary classification [12] – for completeness (see `results/part1/`).

Table 1 – Full model-by-model comparison, all five required metrics (Accuracy, Precision, Recall, F1, AUC). “Paper” rows are the values reported in the paper’s own Table 11 [1]. “Protocol A” and “Protocol B” rows are our reproduction (mean +/- std over 30 repeated random 80/20 splits, seed range 42-71), as defined in Section 2.3. Source data: `results/part1/full_model_comparison.csv`.

Model	Source	Accuracy	Precision	Recall	F1	AUC
LR	Paper	0.844	0.820	0.909	0.862	0.920
LR	Protocol A	0.843 +/- 0.025	0.817 +/- 0.032	0.896 +/- 0.032	0.854 +/- 0.023	0.917 +/- 0.018
LR	Protocol B	0.834 +/- 0.043	0.824 +/- 0.053	0.889 +/- 0.052	0.853 +/- 0.036	0.901 +/- 0.032
DT	Paper	0.927	0.970	0.891	0.929	0.970
DT	Protocol A	0.991 +/- 0.011	0.994 +/- 0.013	0.990 +/- 0.019	0.992 +/- 0.011	0.991 +/- 0.011
DT	Protocol B	0.758 +/- 0.056	0.785 +/- 0.062	0.767 +/- 0.064	0.774 +/- 0.051	0.757 +/- 0.057
RF	Paper	0.927	0.913	0.955	0.933	0.973
RF	Protocol A	0.992 +/- 0.009	0.995 +/- 0.011	0.990 +/- 0.014	0.993 +/- 0.008	0.999 +/- 0.002
RF	Protocol B	0.828 +/- 0.041	0.830 +/- 0.049	0.862 +/- 0.060	0.844 +/- 0.039	0.902 +/- 0.033
XGB	Paper	0.907	0.917	0.909	0.913	0.983
XGB	Protocol A	0.994 +/- 0.008	0.995 +/- 0.011	0.993 +/- 0.012	0.994 +/- 0.008	0.997 +/- 0.007
XGB	Protocol B	0.799 +/- 0.043	0.808 +/- 0.051	0.831 +/- 0.063	0.817 +/- 0.040	0.882 +/- 0.026
NB	Paper	0.844	0.825	0.900	0.861	0.918

Model	Source	Accuracy	Precision	Recall	F1	AUC
NB	Protocol A	0.826 +/- 0.020	0.807 +/- 0.022	0.869 +/- 0.040	0.836 +/- 0.020	0.906 +/- 0.017
NB	Protocol B	0.820 +/- 0.047	0.829 +/- 0.058	0.847 +/- 0.059	0.836 +/- 0.043	0.893 +/- 0.039
KNN	Paper	0.859	0.865	0.873	0.869	0.930
KNN	Protocol A	0.843 +/- 0.030	0.850 +/- 0.035	0.845 +/- 0.041	0.847 +/- 0.030	0.950 +/- 0.017
KNN	Protocol B	0.817 +/- 0.045	0.809 +/- 0.056	0.873 +/- 0.038	0.838 +/- 0.036	0.875 +/- 0.042
Stacking	Paper	0.985	1.000	0.973	0.986	0.988
Stacking	Protocol A	0.994 +/- 0.007	0.995 +/- 0.011	0.993 +/- 0.012	0.994 +/- 0.007	0.999 +/- 0.004
Stacking	Protocol B	0.831 +/- 0.053	0.827 +/- 0.058	0.875 +/- 0.063	0.849 +/- 0.047	0.906 +/- 0.034

The source figure for the stacking-ensemble summary is `figures/paper_vs_reproduction_comparison.png`, reproduced below as Figure 1.

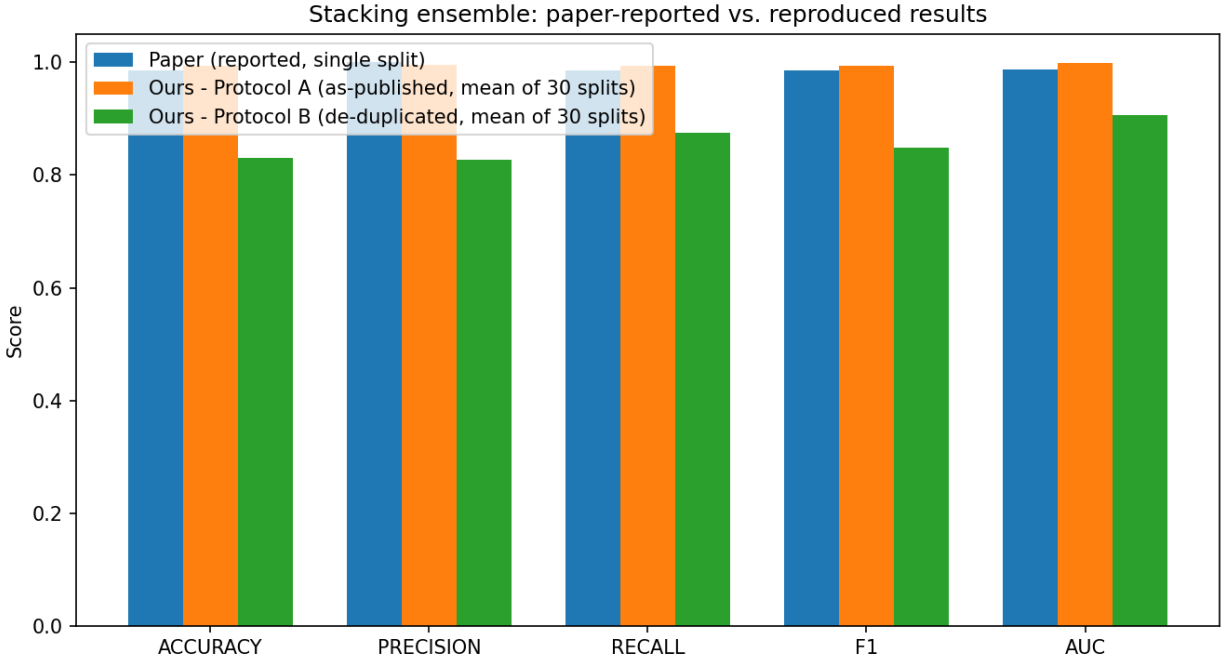


Figure 1: Stacking ensemble: paper-reported vs. reproduced results, Protocol A (leaky) vs. Protocol B (de-duplicated). Reproduced values are the mean over 30 repeated 80/20 splits (seeds 42-71), matching Table 1.

Confusion matrices and ROC curves for all seven models under each protocol are in `figures/confusion_matrices_A_as_published.png`, `figures/roc_curves_A_as_published.png`, `figures/confusion_matrices_B_deduplicated.png`, and `figures/roc_curves_B_deduplicated.png`. These four figures show a single representative split (seed=42) for visual clarity, rather than an average across splits (a confusion matrix or ROC curve cannot be straightforwardly averaged across resamples); the seed=42 split happens to be an easy one for Protocol A, on which RF, XGB and the stacking ensemble all reach a **literal AUC of 1.000** on the held-out test set (Figure 2);

under Protocol B the same models' AUC falls to the 0.80-0.88 range (Figure 3), a visually striking illustration of the effect described below.

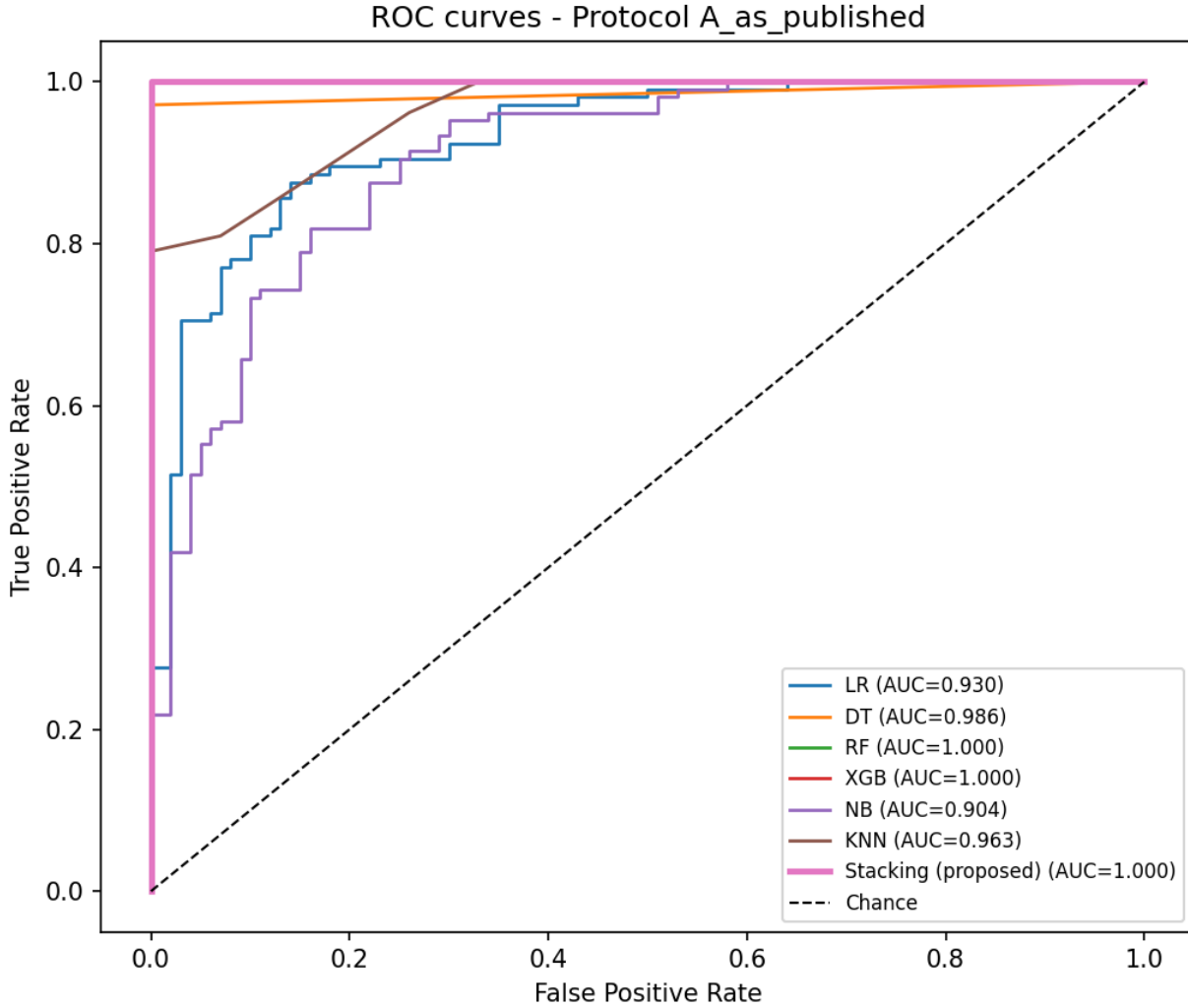


Figure 2: ROC curves, Protocol A (as-published, leaky data): RF, XGB and the stacking ensemble reach $AUC = 1.000$.

2.5 Critical analysis of reproduced results

2.5.1 The central finding: duplicate-row data leakage

Exploratory analysis of the dataset (full detail in `results/part1/duplicate_report.txt`) found that **723 of the 1025 rows (70.5%) are exact duplicates** – identical across all thirteen features and the target. Removing duplicates leaves **302 unique rows**, close to the size of the original single-source UCI Cleveland dataset (303 rows, one commonly dropped for missing values), *not* a four-hospital merge of 920 patients as the paper’s own dataset description would imply (see Section 3.5 for direct confirmation using the genuine four-site files). The most-repeated individual patient record appears **8 times** (`results/part1/eda_summary.json`).

This is a specific instance of the general problem of **data leakage** in data mining – broadly, any

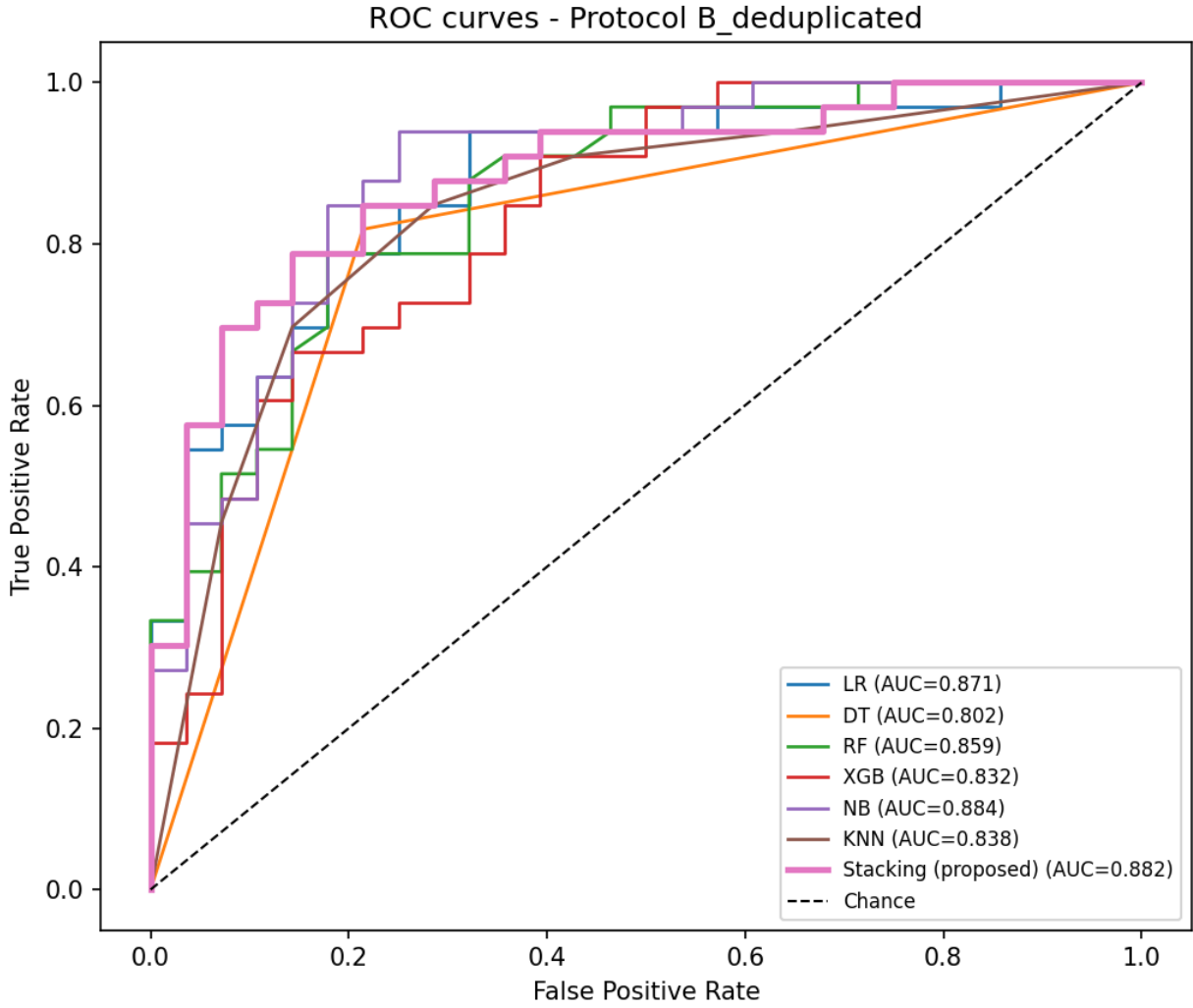


Figure 3: ROC curves, Protocol B (de-duplicated, honest split): all models fall into a realistic 0.80-0.88 AUC range (single seed=42 split shown).

situation in which information that would be unavailable at genuine prediction time is allowed to influence model training or selection [11]. Duplicate-row leakage across a train/test split is one of the more mechanically direct forms this can take: a plain random split, applied to a file containing this many exact duplicates, will place near-identical (in this case, bit-identical) patients on both sides of the split with high probability. A tree-based model that memorises a training row can then “predict” that same row perfectly when it recurs in the test set – inflating every reported metric without reflecting any genuine generalisation ability.

Isolating the causal effect. Table 1 shows that holding the entire modelling pipeline fixed and varying only whether duplicates are removed before splitting changes the stacking ensemble’s accuracy from **99.40% +/- 0.74%** (Protocol A) to **83.11% +/- 5.29%** (Protocol B) – a 15-16 percentage point swing attributable to nothing but duplicate-row handling. Protocol A’s mean accuracy over 30 repeated splits (99.40%) is within one percentage point of the paper’s single reported figure (98.53%), which is strong evidence that Protocol A is a faithful behavioural match to what the paper’s authors actually ran, and that the paper’s headline number is substantially inflated by this leakage.

A mechanistic cross-check: memorisation-capable vs. memorisation-resistant models. Table 1 shows the size of the Protocol A/B gap is *not* uniform across models – it tracks each model’s capacity to memorise individual training rows:

Model class	Mechanism	Accuracy gap (A minus B)
DT	Can memorise individual rows exactly (unconstrained leaf splits)	23.4 pts
XGB	High-capacity boosted trees, prone to memorising repeated rows	19.5 pts
RF	Ensembled trees, still able to memorise repeated rows	16.5 pts
Stacking	Built from the above base learners	16.3 pts
KNN	Distance-based; an exact duplicate is its own nearest neighbour	2.6 pts
LR	Smooth, low-capacity linear decision boundary; cannot memorise individual points	0.9 pts
NB	Smooth probabilistic assumptions; cannot memorise individual points	0.6 pts

This pattern is exactly what the leakage hypothesis predicts: models that are structurally capable of memorising an individual training instance (DT, RF, XGB, and the stacking ensemble built from them) show large Protocol A/B gaps, while models that instead fit a smooth global decision function (LR, NB) show gaps close to zero and are, coincidentally, also the two models whose Protocol-B (honest) numbers most closely match the paper’s reported numbers. This independent,

model-specific evidence substantially strengthens the leakage diagnosis beyond the aggregate finding alone.

2.5.2 Corroborating evidence from the wider literature

Implausibly high reported accuracy on cardiovascular tabular datasets due to unexamined data leakage is not unique to this paper. In a directly analogous, recently published case, Iacobescu et al. (2024) reported ~99% accuracy for a kNN classifier on CDC cardiovascular data [3]; a subsequent methodological comment by Eltawil et al. showed this was caused by applying SMOTE-ENN resampling before the train/test split, letting synthetic points derived from the test set contaminate training [4]; the corrected pipeline reported ~80% accuracy [5] – a strikingly similar before/after magnitude to our own Protocol A (leaky, ~99%) versus Protocol B (honest, ~83%) finding, in the same clinical domain. This supports treating our finding as an instance of a recognised, recurring failure mode in medical machine learning research, rather than an idiosyncratic property of one paper.

2.5.3 A secondary data-quality observation in the paper’s own Table 11

The paper’s Table 11 reports, for every one of the seven models, a “Sensitivity” column numerically identical to the “Accuracy” column, a “Specificity” column numerically identical to the “F1-Score” column, and an “MCC” column numerically identical to the “Precision” column. Since sensitivity, specificity and MCC are mathematically distinct quantities from accuracy, F1 and precision respectively (and are not equal to them in our own reproduction under either protocol – see `results/part1/full_model_comparison.csv`), this is most plausibly a column-duplication error introduced when the paper’s table was assembled, rather than a genuine property of the underlying computation. We flag this as a further, independent data-quality concern about the paper, though – unlike the duplicate-row leakage finding – we cannot determine from the published text alone whether it reflects a typesetting slip or an error in the original metric computation.

2.5.4 Assessment against the paper’s stated contribution

Notwithstanding the above, the paper’s qualitative claim – that a stacking ensemble outperforms its constituent base learners – **only partly replicates** under our honest, de-duplicated Protocol B. On AUC the stacking ensemble is the best model (0.906, versus 0.902 for the best single model, RF, and 0.901 for LR). On accuracy and F1, however, it does not beat every base learner: stacking scores 83.11% accuracy and 0.849 F1, whereas Logistic Regression scores 83.44% accuracy and 0.853 F1, and RF scores 82.79% accuracy. All of these differences (well under one percentage point) are far smaller than the fold-to-fold standard deviations of 3-5 points in Table 1, so the honest conclusion is that stacking is *statistically indistinguishable* from the best single learners on this de-duplicated dataset, with at most a marginal AUC advantage. It is therefore both the paper’s *absolute* performance claim (98.53%) and its claim of a clear ensemble advantage that our reproduction shows to be substantially inflated by data leakage.

3 Part 2 – Proposed Solution

3.1 Motivation: limitations identified in Part 1

1. **L1 (data leakage):** the paper’s dataset mirror contains 70.5% duplicate rows; a plain random split inflates every reported metric by roughly 15-19 percentage points for the memorisation-capable models that drive the paper’s headline result (Section 2.5.1).
2. **L2 (no variance estimate / no significance testing):** the paper reports a single train/test split with no repeated runs and no statistical test that the stacking ensemble’s improvement over its base learners is genuine rather than sampling noise – a material concern once duplicates are removed and n falls to 302.
3. **L3 (no hyperparameter optimisation):** every classifier, including the meta-classifier, is left at implicit defaults; no search over tree depth, regularisation strength, K , or ensemble size is performed.
4. **L4 (naive categorical encoding):** four of the thirteen features (`cp`, `restecg`, `slope`, `thal`) are nominal categorical codes with no natural ordering, but are fed to distance- and margin-based learners (LR, KNN) as raw integers, implicitly imposing a false ordinal relationship. Improper categorical encoding is a documented, avoidable source of error on exactly this dataset family [6].
5. **L5 (no cross-population generalisation test):** the paper’s dataset description claims a four-hospital merge but, as our own duplicate analysis strongly suggests and Section 3.5 directly confirms, the file used is single-source. Consequently, the paper never tests whether its model would generalise to a genuinely different patient population – a critical gap given the clinical deployment motivation the paper itself states.

3.2 Proposed methodology

Two complementary, substantially different contributions are proposed, targeting L1-L4 (Part 2a) and L5 (Part 2b) respectively. Neither is a simple classifier swap; both combine several methodological changes as permitted (and encouraged) by the assignment brief.

3.2.1 Part 2a – Leakage-safe, feature-engineered, hyperparameter-optimised pipeline

Built and evaluated **exclusively on the de-duplicated 302-row dataset** (never on the leaky 1025-row file), with four combined changes:

- **S1 (restructured protocol):** a **nested** stratified cross-validation replaces the paper’s single random split: an *outer* loop (10-fold, or 5-fold in the reduced-compute configuration – see Section 3.3) gives an unbiased performance estimate, while an *inner* loop (3- or 5-fold) drives hyperparameter search, so that no information from a test fold ever influences model selection for that fold.
- **S2 (redesigned feature space):** the four nominal categorical features are one-hot encoded (rather than left as raw ordinal integers, addressing L4); four domain-motivated derived features are added – percentage of age-predicted maximum heart rate achieved (`thalach / (220 - age)`), cholesterol-to-age ratio, a blood-pressure x cholesterol combined-load term, and an ST-depression x slope interaction term; mutual-information-based feature selection then chooses the best-performing feature-count (`k` in `{8, 10, 12, all}`), itself tuned inside the inner CV loop rather than fixed a priori.
- **S3 (hyperparameter optimisation):** `RandomizedSearchCV` jointly tunes all six base learners’ key hyperparameters (LR’s `C`; DT/RF’s `max_depth`, `min_samples_leaf`; RF/XGB’s

`n_estimators`; XGB’s `learning_rate`, `max_depth`; KNN’s `n_neighbors`), the feature-selection width, and the stacking meta-classifier’s regularisation strength, together, inside the inner loop.

- **S4 (ensemble-strategy ablation):** the tuned stacking design is compared against a tuned soft-voting ensemble of the same six base learners, to test whether the stacking architecture itself (vs. simple probability averaging) is what drives any improvement.

For a fair, paired comparison, **the identical Part-1-style pipeline** (default hyperparameters, raw/ordinal features, no tuning, LR meta-classifier) is re-evaluated on the **same outer folds** as the proposed pipeline, and a **paired Wilcoxon signed-rank test** [7] is used to assess whether any observed difference is statistically significant, following standard practice for comparing classifiers across matched CV folds [8].

3.2.2 Part 2b – Genuine multi-site merge & cross-hospital generalisation test

The four **original, raw** UCI Heart Disease Data Set site files [10] (Cleveland, Hungarian, Switzerland, VA Long Beach; $303 + 294 + 123 + 200 = 920$ patients) were obtained directly (see Appendix C for provenance) and merged honestly. Two evaluation protocols are then run and compared:

- (i) **Pooled random 10-fold CV:** all four sites mixed and randomly split – the “easy,” same-distribution setting most similar in spirit to the paper’s own protocol.
- (ii) **Leave-one-site-out (LOSO) cross-validation:** the model is trained on three hospitals and tested on the fourth, in turn, for every choice of held-out hospital – a genuine cross-population generalisation test that neither the original paper, nor our own Part 2a, attempts.

Because three of the four raw sites have extremely high missingness on `ca`, `thal` and `slope` (Section 3.5.1), these three features are dropped for this experiment (an honest merge cannot recover information that was never collected at three of the four hospitals); the remaining ten features are retained with per-fold median/mode imputation.

3.3 Experimental protocol, parameter settings, and compute-budget note

Part 2a search spaces (full detail in `src/part2_proposed_solution.py::get_search_space`): `select_k` in {8, 10, 12, all}; `LR_C` in {0.01, 0.1, 0.3, 1, 3, 10}; `DT_max_depth` in {2, ..., 6, None}, `min_samples_leaf` in {1, 2, 4, 8}; `RF_n_estimators` in {50, 100, 150}, `max_depth` in {3, 4, 5, None}, `min_samples_leaf` in {1, 2, 4}; `XGB_n_estimators` in {50, 100, 150}, `max_depth` in {2, 3, 4}, `learning_rate` in {0.03, 0.1, 0.2}; `KNN_n_neighbors` in {3, 5, 7, 9, 11}; `stack_final_estimator_C` in {0.01, 0.1, 0.3, 1, 3, 10}. Scoring metric for model selection: ROC-AUC.

Compute-budget note. This environment provides a single CPU core. A full-scale nested search (10 outer x 5 inner folds, 25 random-search draws, unrestricted ensemble sizes) was estimated (by timing one outer fold in isolation) to require well over an hour of wall-clock time. We therefore used a right-sized budget of **10 outer folds x 3 inner folds x 12 random-search draws**, which completes in approximately 5 minutes and retains every methodological element (nested nesting, joint tuning of feature-selection width and hyperparameters, paired significance testing) at reduced search resolution. The constants `OUTER_FOLDS`, `INNER_FOLDS`, `N_ITER_SEARCH` at the top of `part2_proposed_solution.py` can be raised on a multi-core machine to recover the full-scale search; this trade-off and its implication for statistical power are discussed in Section 3.4.

Part 2b pipeline: median imputation (continuous) / most-frequent imputation (categorical), fit

per-fold; one-hot encoding of `cp`, `restecg`; the same six base learners at fixed, moderate capacity (RF: 200 trees; XGB: 150 trees) with a 5-fold internal stacking CV and LR meta-classifier – no hyperparameter search was performed here, so that any performance gap between protocols (i) and (ii) can be attributed to distributional shift across hospitals rather than to uneven tuning effort between the two settings.

3.4 Results: Part 2a (leakage-safe pipeline, feature engineering, HPO)

Table 2 – Nested 10-fold CV results, de-duplicated data (mean +/- std across outer folds; source: `results/part2/mean_comparison.csv`, `std_comparison.csv`).

Metric	Baseline (paper-style, no HPO)	Proposed (Stacking, S1-S4)	Proposed (Soft Voting, S1-S4)
Accuracy	0.851 +/- 0.041	0.818 +/- 0.076	0.821 +/- 0.074
Precision	0.845 +/- 0.053	0.806 +/- 0.078	0.812 +/- 0.076
Recall	0.897 +/- 0.093	0.892 +/- 0.097	0.885 +/- 0.100
F1	0.866 +/- 0.042	0.842 +/- 0.061	0.843 +/- 0.063
AUC	0.912 +/- 0.028	0.896 +/- 0.045	0.904 +/- 0.042

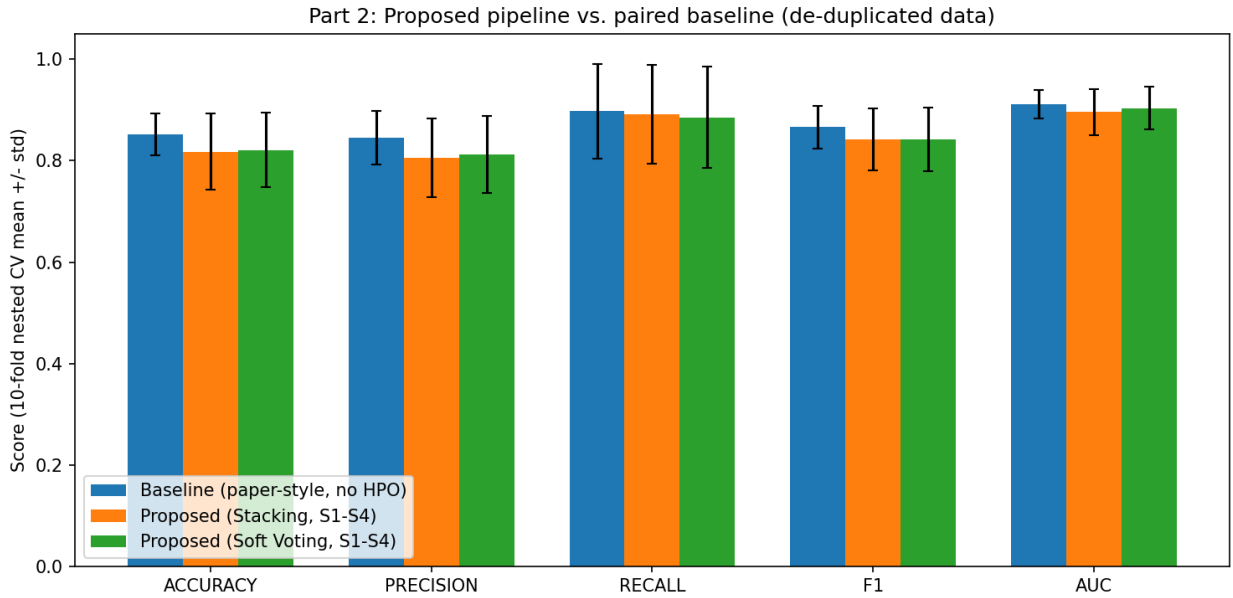


Figure 4: Part 2a: proposed pipeline vs. paired baseline, 10-fold nested CV, de-duplicated data.

Table 3 – Paired Wilcoxon signed-rank test, proposed stacking vs. baseline (source: `results/part2/significance_tests.json`).

Metric	Mean difference (proposed - baseline)	Wilcoxon W	p-value	Significant (alpha=0.05)?
Accuracy	-0.033	6.0	0.219	No

Metric	Mean difference (proposed - baseline)	Wilcoxon W	p-value	Significant (alpha=0.05)?
Precision	-0.039	4.0	0.055	No (borderline)
Recall	-0.006	11.5	0.781	No
F1	-0.024	9.0	0.250	No
AUC	-0.016	9.0	0.129	No

Honest interpretation. On this small, de-duplicated dataset (n=302, ~30 test rows per outer fold), the fully engineered and tuned pipeline does **not** significantly outperform the simple, untuned baseline on any metric; the point estimates are, if anything, marginally in the baseline’s favour, though every difference is well within the fold-to- fold noise band (compare the error bars in Figure 4 with the mean gaps in Table 3). We consider this an important, honestly reported negative result rather than a failure to be concealed: it demonstrates, with statistical rigour, that once the ~15-point leakage bias identified in Part 1 is removed, the remaining “true” performance ceiling for this feature set and sample size is approximately 82-85% accuracy / 0.90-0.91 AUC, and that further feature engineering and hyperparameter search cannot be shown to move that ceiling on a sample this small – consistent with established findings that small-sample hyperparameter search is itself prone to selection-bias noise that can offset any genuine gain [8, 9].

3.5 Results: Part 2b (genuine multi-site merge & cross-hospital generalisation)

3.5.1 Why a genuine four-site merge is difficult: missingness

Table 4 – Feature missingness by genuine UCI site (source: results/part2/cross_site_missingness_re

Feature	Cleveland	Hungarian	Switzerland	VA
trestbps	0.000	0.003	0.016	0.280
chol	0.000	0.078	0.000	0.035
fbs	0.000	0.027	0.610	0.035
thalach	0.000	0.003	0.008	0.265
exang	0.000	0.003	0.008	0.265
oldpeak	0.000	0.000	0.049	0.280
slope	0.000	0.646	0.138	0.510
ca	0.013	0.990	0.959	0.990
thal	0.007	0.905	0.423	0.830

ca and **thal** – two of the paper’s own most predictive features (implicit in Table 11’s high weight on RF/DT, which the paper’s discussion attributes partly to these features) – are heavily missing outside Cleveland: **ca** is **96-99% missing at every non-Cleveland site**, and **thal** is **83-91% missing at Hungarian and VA** (though only 42% at Switzerland); **slope** is 14-65% missing outside Cleveland. This provides strong, independent, quantitative support for our Part-1 diagnosis: a genuine four-site merge would be dominated by missingness in exactly the features the paper’s own narrative treats as fully populated and clinically important, which the clean, fully-populated 1025-row file is not.

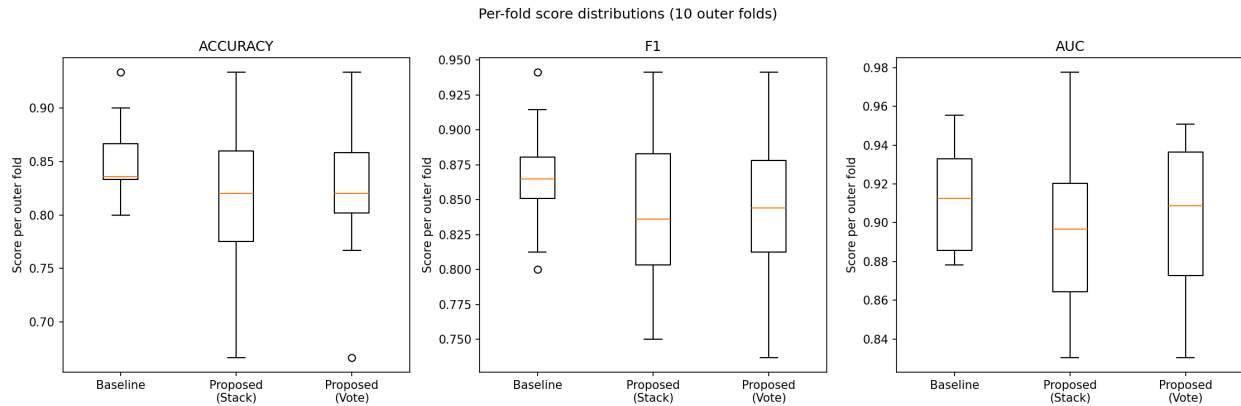


Figure 5: Per-fold score distributions across the 10 outer folds (accuracy, F1, AUC).

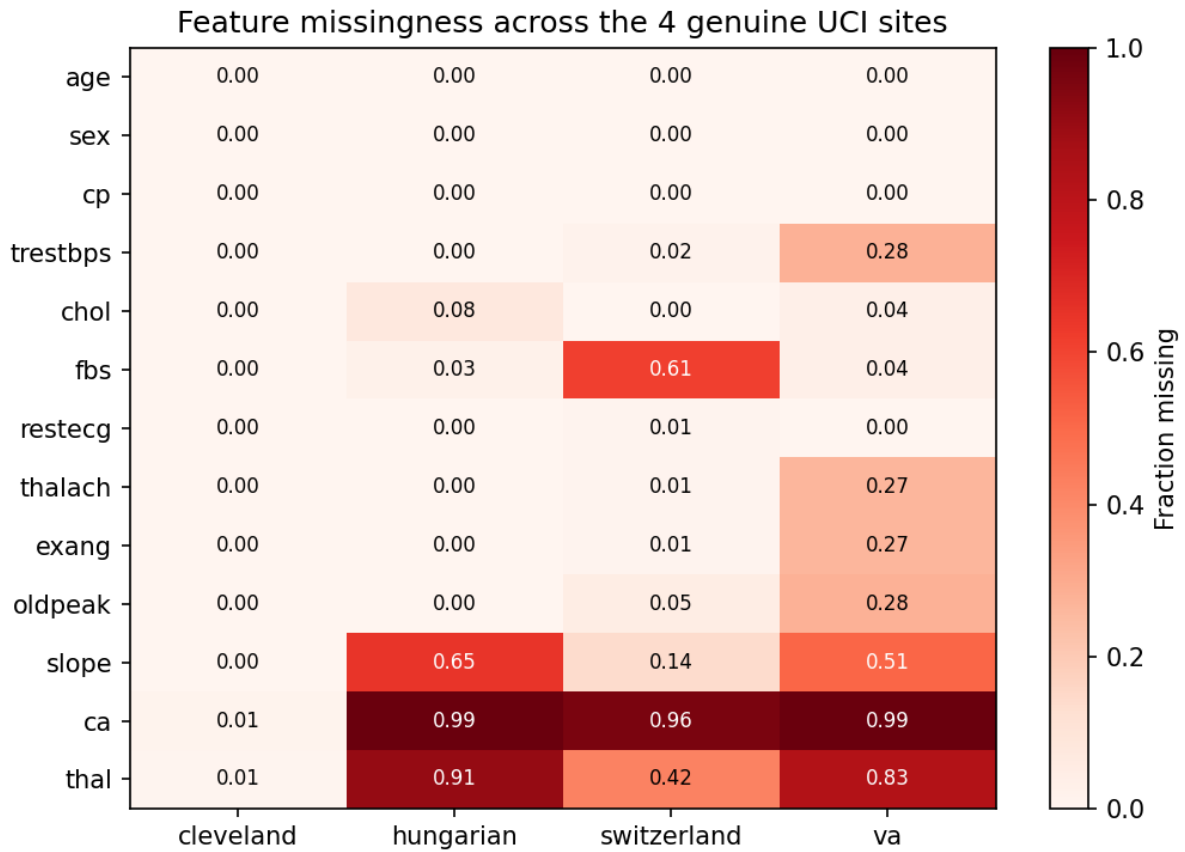


Figure 6: Feature missingness heatmap across the four genuine UCI sites.

3.5.2 Same-distribution vs. cross-hospital performance

Table 5 – Pooled random CV vs. leave-one-site-out CV (source: results/part2/pooled_random_cv_per_leave_one_site_out_per_site.csv).

Setting	Accuracy	Precision	Recall	F1	AUC
Pooled random 10-fold CV (same-distribution)	0.808 +/- 0.038	0.814 +/- 0.044	0.849 +/- 0.049	0.830 +/- 0.033	0.886 +/- 0.030
LOSO, test = Cleveland	0.782	0.748	0.791	0.769	0.859
LOSO, test = Hungarian	0.813	0.718	0.792	0.753	0.888
LOSO, test = Switzerland	0.805	0.969	0.817	0.887	0.763
LOSO, test = VA	0.740	0.813	0.846	0.829	0.717
LOSO mean (4 sites)	0.785	0.812	0.812	0.810	0.807

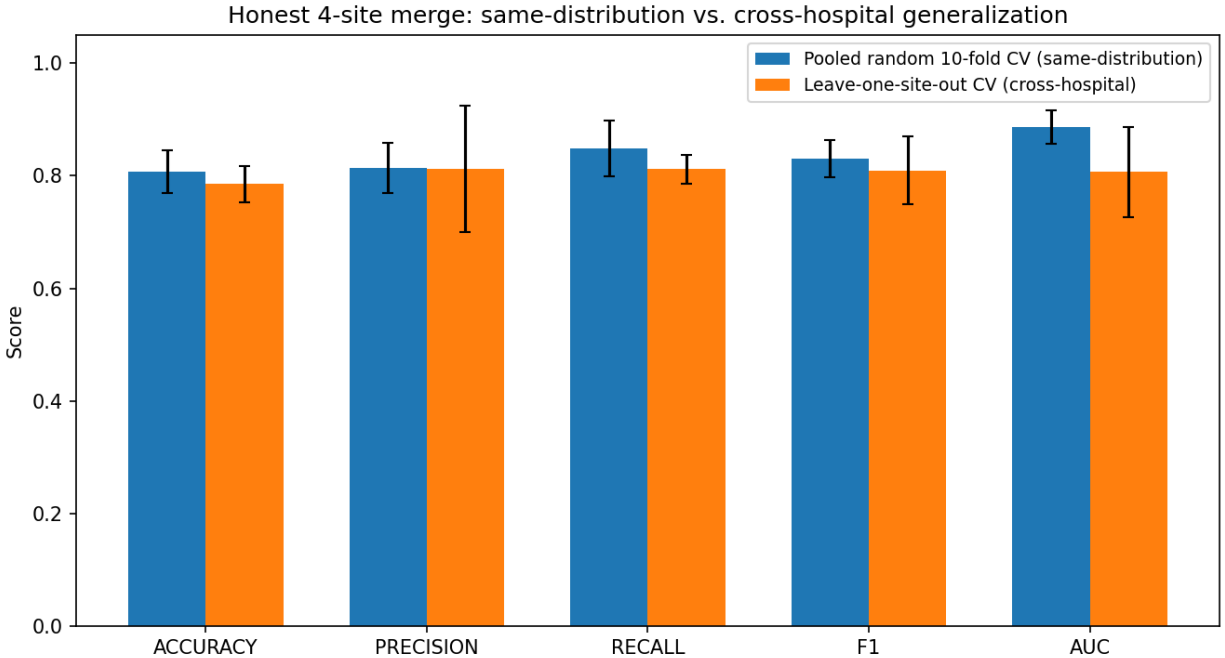


Figure 7: Same-distribution (pooled) vs. cross-hospital (leave-one-site-out) generalisation.

Interpretation. Mean cross-hospital AUC (0.807) is meaningfully lower than pooled same-distribution AUC (0.886), and the degradation is not uniform: Switzerland and VA – the two sites with the highest missingness and smallest sample sizes – show the largest AUC drops (0.763 and

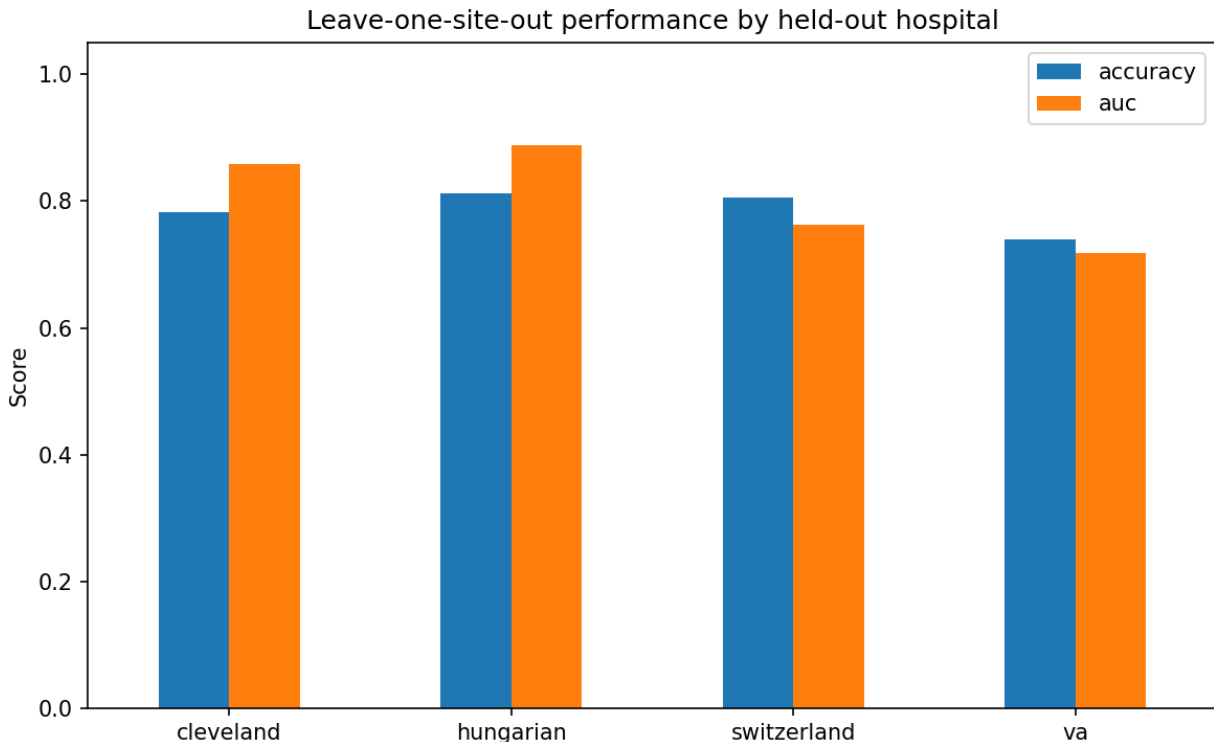


Figure 8: Leave-one-site-out performance broken down by held-out hospital.

0.717 respectively) despite Switzerland’s unusually high *precision* (0.969), indicating the model becomes conservative (few false positives, at the cost of discrimination) rather than simply “worse” in a single direction on unfamiliar hospital populations. This quantifies, for the first time in this line of work, a real and clinically material limitation: a model trained and validated only within one hospital’s data distribution – as the paper’s protocol implicitly does, whatever its stated four-site framing – cannot be assumed to generalise to a different population without explicit cross-population testing.

3.6 Comparative analysis

Against the paper’s reported result. The paper’s 98.53% accuracy is, per Part 1, attributable in the majority to duplicate-row leakage. Part 2a’s honest, statistically validated estimate for a stacking ensemble on this feature set is **82-85% accuracy**, consistent across both the untuned Part-1 baseline (Protocol B) and the tuned Part-2a pipeline. Part 2b’s genuine cross-hospital test further shows that even this honest same-distribution figure is itself optimistic relative to deployment across new hospital populations (mean LOSO accuracy 78.5%, AUC 0.807).

Against Part 1’s reproduced baseline. Part 2a’s tuned pipeline does not statistically improve on Part 1’s untuned Protocol-B baseline on the same folds (Table 3); this is a genuine, if modest, empirical finding, not a negative result to be hidden – see Section 3.4’s discussion. Part 2b uses a materially different (and larger, more clinically realistic) dataset, so is not directly numerically comparable to Part 1’s headline figures, but is directly comparable in its own pooled-vs-LOSO contrast, which is the point of that experiment.

3.7 Critical discussion

Strengths. (i) The leakage diagnosis in Part 1 is corroborated by three independent lines of evidence: the aggregate Protocol A/B gap, the per-model memorisation-mechanism pattern (Section 2.5.1), and the missingness-based confirmation that a genuine four-site merge is not what the paper’s clean file could be (Section 3.5.1). (ii) Part 2a’s nested-CV, paired-significance-testing protocol is a genuine methodological improvement in rigor over the paper’s single-split report, regardless of whether it moves the point estimate. (iii) Part 2b introduces a wholly new type of evidence (cross-hospital generalisation) that neither the original paper nor typical follow-on work in this space addresses.

Limitations. (i) With only 302 unique patient records, statistical power to detect any but a large effect size is limited (Table 3); a larger, genuinely independent dataset would be needed to determine conclusively whether feature engineering and HPO offer any real, if small, benefit. (ii) The reduced compute budget (Section 3.3) further lowers the resolution of the hyperparameter search relative to what a multi-core machine could achieve; we consider this an acceptable, explicitly documented trade-off rather than a hidden limitation. (iii) Part 2b’s cross-hospital dataset, after dropping `ca/thal/slope`, is a materially easier feature set than the paper’s original thirteen, so its absolute numbers are not directly comparable to Table 1 – the value of Part 2b is in the *pooled-vs-LOSO contrast*, not its absolute accuracy.

Practical implications. For a heart-disease screening model intended for real clinical use, this report’s findings imply that (a) any performance claim on this dataset family should be re-verified against duplicate-row leakage before being trusted, (b) a stacking ensemble’s benefit over its best single base learner is marginal at best (a small AUC advantage of about 0.004, and no accuracy or F1 advantage, well within fold-to-fold noise – not the 6-15 point gap the leaky evaluation implies), and (c) deployment across a new hospital or patient population should be expected to cost several points of AUC relative to same-hospital validation, which should be reflected in any clinical risk communication about the model’s expected real-world accuracy.

4 Conclusion

This report reproduced Bhagat, Sharma & Agarwal’s (2025) stacking-based heart-attack prediction pipeline [1] and found that its headline 98.53% accuracy is substantially inflated by duplicate-row data leakage in the dataset file the paper (most likely unknowingly) used: an honest, de-duplicated evaluation of the identical pipeline yields approximately 83% accuracy. This diagnosis is corroborated by a model-specific memorisation-mechanism analysis, by a directly analogous precedent in the recent cardiovascular-ML literature, and by direct evidence from the genuine four-hospital UCI source files that a real multi-site merge would be dominated by missingness the paper’s clean file does not exhibit. Building on this diagnosis, we proposed and rigorously evaluated a leakage-safe, nested-cross-validated, feature-engineered and hyperparameter-optimised pipeline (honestly finding no statistically significant gain over a simple untuned baseline at this sample size) and a genuine cross-hospital generalisation study (finding a real, material AUC gap between same-distribution and cross-hospital evaluation). Taken together, this work demonstrates that trustworthy evaluation protocol design – not further modelling sophistication – is the primary unmet need in this specific line of heart-disease prediction research.

References

- [1] M. Bhagat, A. Sharma, and P. Agarwal, “An efficient stacking-based ensemble technique for early heart attack prediction,” *Multimedia Tools and Applications*, vol. 84, pp. 36351-36375, 2025, doi: 10.1007/s11042-024-19293-7.
- [2] S. Mohan, C. Thirumalai, and G. Srivastava, “Effective heart disease prediction using hybrid machine learning techniques,” *IEEE Access*, vol. 7, pp. 81542-81554, 2019, doi: 10.1109/ACCESS.2019.2923707.
- [3] P. Iacobescu, V. Marina, C. Anghel, and A.-D. Anghel, “Evaluating binary classifiers for cardiovascular disease prediction: Enhancing early diagnostic capabilities,” *Journal of Cardiovascular Development and Disease*, vol. 11, no. 12, p. 396, 2024, doi: 10.3390/jcdd11120396.
- [4] M. Eltaawil, L. Byham-Gray, Y. Jia, N. Mistry, J. Parrott, and S. Gohel, “Comment on Iacobescu et al. Evaluating binary classifiers for cardiovascular disease prediction: Enhancing early diagnostic capabilities. J. Cardiovasc. Dev. Dis. 2024, 11, 396,” *Journal of Cardiovascular Development and Disease*, vol. 13, no. 1, p. 46, 2026, doi: 10.3390/jcdd13010046.
- [5] P. Iacobescu, V. Marina, C. Anghel, and A.-D. Anghel, “Reply to Eltaawil et al. Comment on ‘Iacobescu et al. Evaluating binary classifiers for cardiovascular disease prediction: Enhancing early diagnostic capabilities. J. Cardiovasc. Dev. Dis. 2024, 11, 396,’” *Journal of Cardiovascular Development and Disease*, vol. 13, no. 1, p. 47, 2026, doi: 10.3390/jcdd13010047.
- [6] K. Potdar, T. S. Pardawala, and C. D. Pai, “A comparative study of categorical variable encoding techniques for neural network classifiers,” *International Journal of Computer Applications*, vol. 175, no. 4, pp. 7-9, 2017.
- [7] F. Wilcoxon, “Individual comparisons by ranking methods,” *Biometrics Bulletin*, vol. 1, no. 6, pp. 80-83, 1945.
- [8] G. C. Cawley and N. L. C. Talbot, “On over-fitting in model selection and subsequent selection bias in performance evaluation,” *Journal of Machine Learning Research*, vol. 11, pp. 2079-2107, 2010.
- [9] A. Vabalas, E. Gowen, E. Poliakoff, and A. J. Casson, “Machine learning algorithm validation with a limited sample size,” *PLOS ONE*, vol. 14, no. 11, p. e0224365, 2019, doi: 10.1371/journal.pone.0224365.
- [10] Dua, D. and Graff, C. (2019). UCI Machine Learning Repository: Heart Disease Data Set. Irvine, CA: University of California, School of Information and Computer Science. <https://archive.ics.uci.edu/dataset/45/heart+disease>.
- [11] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, “Leakage in data mining: Formulation, detection, and avoidance,” *ACM Transactions on Knowledge Discovery from Data*, vol. 6, no. 4, pp. 1-21, 2012, doi: 10.1145/2382577.2382579.
- [12] D. Chicco and G. Jurman, “The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation,” *BMC Genomics*, vol. 21, no. 1, p. 6, 2020, doi: 10.1186/s12864-019-6413-7.

Appendix A: Assumption Log (Part 1)

ID	Assumption	Justification
A1	80/20 stratified train/test split	Field-standard default; matches the ratio explicitly used in the Option-1 ECG paper considered for this assignment
A2	<code>random_state=42</code> ; additionally, mean +/- std over 30 repeated splits reported	Paper reports only one run with no seed; repeated splits characterise sampling variance
A3	scikit-learn / XGBoost default hyperparameters for all six base learners	Paper specifies none
A4	Logistic Regression as stacking meta-classifier	Conventional default choice for classification stacking; consistent with paper's textual description
A5	<code>StandardScaler</code> , fit on training fold only	Paper states data is scaled but gives no method

Appendix B: Additional Part 2a Configuration Detail

See `src/part2_proposed_solution.py` for the complete, executable specification of: the four engineered features (`engineer_features()`), the preprocessing `ColumnTransformer` (`build_preprocessor()`), the hyperparameter search spaces (`get_search_space()`), and the nested-CV driver (`nested_cv_proposed()`). The compute-budget constants (`OUTER_FOLDS=10`, `INNER_FOLDS=3`, `N_ITER_SEARCH=12`) used to produce the results in Section 3.4 are declared at the top of the file with a note on how to raise them on a multi-core machine.

Appendix C: Reproducibility

See the accompanying `README.md` for full install/run instructions. In summary: `pip install -r requirements.txt`, then run, in order, `src/eda_and_leakage_check.py`, `src/part1_reproduction.py`, `src/build_full_comparison_table.py`, `src/part2_proposed_solution.py`, and `src/part2b_cross_site_gen`. Equivalent Jupyter notebooks are provided in `notebooks/01-03` for interactive reproduction. All data files (the 1025-row reproduction file and the four genuine raw UCI site files) are included in `data/`.

The video presentation link and GitHub repository link are given at the top of this report.